

S.No	Component	Quantity	Purpose	Operating Voltage	Approx Weight
1	ESP32 DevKit V1 (30-pin, ESP32-WROOM-32)	1	Main microcontroller, WiFi communication	3.3V logic / 5V VIN	10–12 g
2	BME280	1	Measures temperature, humidity & pressure	3.3V	~2 g
3	MQ-135 Gas Sensor	1	Detects air pollutants and harmful gases	5V	8–10 g
4	DHT11 Sensor Module	1	Additional temperature & humidity sensing (optional)	3.3V–5V	~3 g
5	3.7V Li-Po Battery (1000mAh)	1	Lightweight power source	3.7V	20–25 g
6	TP4056 Charging Module (with protection)	1	Safe Li-Po charging and protection	5V USB input	~3 g
7	MT3608 Boost Converter	1	Steps up 3.7V battery to 5V for MQ135	2V–24V input	~5 g
8	ON/OFF Mini Switch	1	Main power control	—	~2 g
9	Breadboard / Mini PCB	1	Temporary or permanent circuit mounting	—	10–15 g
10	Jumper Wires	Multiple	Electrical connections	—	~5 g
11	Foam / Thermocol Payload Box	1	Lightweight enclosure for payload	—	15–20 g
12	Nylon Thread	1	Tethering balloon and payload	—	Negligible

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13	Latex Helium Balloon	1	Lifts payload into air	—	Depends on size
14	Helium Gas	As required	Balloon lifting gas	—	—
15	Micro USB Cable	1	Programming ESP32	5V	—
16	Electrical Tape / Zip Ties	Few	Securing wiring and payload	—	Negligible
17	Capacitor (220μF–470μF recommended)	1	Stabilizes MQ135 power supply	5V+	Negligible